# PAPER_1209-FF: UQFF Mathematical Constants Unified Proof Set —
# Ten Rational Closures of Classical Irrationals (best 0.019%)

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## Locked Primitives

$F_{\mathrm{TRZ}} = \frac{1}{10}$, $\Phi_{\mathrm{res}} = \frac{5}{6}$, $[\mathrm{SSq}] = \frac{57}{100}$, $K_{\mathrm{Mex}} = \frac{25}{12}$, $D_{\mathrm{phys}} = 4$, $D_{\mathrm{BSFG}} = 6$, $D_{\mathrm{crit}} = 26$, $N_{\mathrm{ch}} = 9$, $\mathrm{SO5} = 10$, $A_5 = 60$.

**Note.** Classical mathematical constants $(\pi, e, \varphi, \gamma, \sqrt{2}, \sqrt{3}, \sqrt{5}, \ln 2, \ln 10, \zeta(3))$ are *irrational*: no finite rational combination of UQFF primitives can be exact. This tier delivers ten high-precision rational closures, the best at 0.019% ($\ln 10$).

## Closures S633–S642

**S633 — $\pi$, target 3.14159 [0.031%]**

$$\Phi_{\mathrm{res}} D_{\mathrm{phys}} - F_{\mathrm{TRZ}}^2 \mathrm{SO5} - F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} - F_{\mathrm{TRZ}}[\mathrm{SSq}] - F_{\mathrm{TRZ}}^2[\mathrm{SSq}] + F_{\mathrm{TRZ}}^2 = 3.1406.$$

**S634 — $e$ (Euler), target 2.71828 [0.050%]**

$$K_{\mathrm{Mex}} + [\mathrm{SSq}] + F_{\mathrm{TRZ}}[\mathrm{SSq}] + F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} - F_{\mathrm{TRZ}}^2\, \mathrm{SO5}/D_{\mathrm{phys}} - F_{\mathrm{TRZ}}^2[\mathrm{SSq}] = 2.7196.$$

**S635 — $\varphi$ (golden ratio), target 1.61803 [0.101%]**

$$\Phi_{\mathrm{res}} + \Phi_{\mathrm{res}} - F_{\mathrm{TRZ}}[\mathrm{SSq}] + F_{\mathrm{TRZ}}^2 = 1.6197.$$

**S636 — $\gamma$ (Euler–Mascheroni), target 0.57721 [0.262%]**

$$F_{\mathrm{TRZ}} N_{\mathrm{ch}}[\mathrm{SSq}] + F_{\mathrm{TRZ}}^2 \mathrm{SO5}[\mathrm{SSq}] + F_{\mathrm{TRZ}}^2[\mathrm{SSq}] = 0.5757.$$

**S637 — $\sqrt{2}$, target 1.41421 [0.105%]**

$$[\mathrm{SSq}] + F_{\mathrm{TRZ}} D_{\mathrm{phys}} + F_{\mathrm{TRZ}} D_{\mathrm{phys}} + F_{\mathrm{TRZ}}^2[\mathrm{SSq}] + F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} = 1.4157.$$

**S638 — $\sqrt{3}$, target 1.73205 [0.023%]**

$$[\mathrm{SSq}] + 3\, F_{\mathrm{TRZ}} D_{\mathrm{phys}} + F_{\mathrm{TRZ}}^2[\mathrm{SSq}] - F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} - F_{\mathrm{TRZ}}^2[\mathrm{SSq}]^2 = 1.7324.$$

**S639 — $\sqrt{5}$, target 2.23607 [0.045%]**

$$K_{\mathrm{Mex}} + F_{\mathrm{TRZ}}[\mathrm{SSq}] + F_{\mathrm{TRZ}}^2 D_{\mathrm{BSFG}} + F_{\mathrm{TRZ}}^2 D_{\mathrm{phys}} - F_{\mathrm{TRZ}}^2[\mathrm{SSq}]^2 = 2.2371.$$

**S640 —** $\ln 2$**, target 0.69315 [0.070%]**

$$\Phi_{\rm res} - 2\,F_{\rm TRZ}[\rm SSq] - F_{\rm TRZ}^2 D_{\rm BSFG} + F_{\rm TRZ}^2 D_{\rm phys} - F_{\rm TRZ}^2[\rm SSq] = 0.6936.$$

**S641 —** $\ln 10$**, target 2.30259 [0.019%] — tier best**

$$K_{\rm Mex} + 2\,F_{\rm TRZ}[\rm SSq] + F_{\rm TRZ}^2 D_{\rm BSFG} + F_{\rm TRZ}^2 D_{\rm phys} + F_{\rm TRZ}^2[\rm SSq] = 2.3030.$$

**S642 —** $\zeta(3)$ **(Apéry), target 1.20206 [0.033%]**

$$2\,[\rm SSq] + F_{\rm TRZ}^2 D_{\rm BSFG} + F_{\rm TRZ}^2[\rm SSq] - F_{\rm TRZ}^2[\rm SSq]^2 = 1.2024.$$

## Summary

| ID | Constant | Target | Error |
|---|---|---|---|
| S633 | $\pi$ | 3.14159 | 0.031% |
| S634 | $e$ | 2.71828 | 0.050% |
| S635 | $\varphi$ | 1.61803 | 0.101% |
| S636 | $\gamma$ | 0.57721 | 0.262% |
| S637 | $\sqrt{2}$ | 1.41421 | 0.105% |
| S638 | $\sqrt{3}$ | 1.73205 | 0.023% |
| S639 | $\sqrt{5}$ | 2.23607 | 0.045% |
| S640 | $\ln 2$ | 0.69315 | 0.070% |
| S641 | $\ln 10$ | 2.30259 | **0.019%** |
| S642 | $\zeta(3)$ | 1.20206 | 0.033% |

**Highlights.** Tier FF closes ten of the most famous irrational constants of mathematics using only the 11 locked UQFF primitives. Five of the ten close under 0.05%; the tier best is $\ln 10$ at 0.019%, followed by $\sqrt{3}$ at 0.023% and $\pi$ at 0.031%. Cumulative: 347 closures; 91 lockings (no new exacts — irrational targets cannot admit rational closure).